



Pietro Ferraiuolo

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Gender: Male **Date of birth**: 7 Oct 1996 **Nationality**: Italian

ABOUT ME

I graduated in *Astronomy and Astrophysics* from 'La Sapienza - University of Rome' driven by a fervent desire to become a scientific researcher. I am dedicated to a continuous self-improvement, with strong proficiency in technology and computer programming. Beyond my academic pursuits, I'm an enthusiastic traveler who enjoys exploring new cultures. In my leisure time, I like to play video games, to read, and I love engaging in board games with my friends, as cherished hobbies.

CURRENT OCCUPATION

[15 Apr 2024 – Current] **Lead Software Developer for M4**

Research Fellowship: **Optical Qualification of the Test Tower for the ELT's M4 Adaptive Mirror and the DP Prototype.**

HONOURS AND AWARDS

[Feb 2025] **Grant di Ricerca Fondamentale Awarding institution**: INAF - Istituto Nazionale di Astrofisica

Awarded funding for the fundamental Astrophysics Call 2024. PI of the **BINCAT** project.

ONGOING RESEARCH

[Feb 2025 – Current] **BINCAT - Binary Catalog for AO Sytems Performace Evaluation**

Principal Investigator, INAF Fundamental Astrophysics Call 2024 (miniGrant Awardee)

Awarded funding for the development of a catalog of close binary stars (20–200 mas) as a tool for commissioning and optimizing adaptive optics (AO) systems on large telescopes. The project utilizes GAIA DR3 data to statistically identify multiplicity indicators and optimize the parameter space through PSF simulations. The catalog will support both current and next-generation AO instruments (e.g., MAVIS, MORFEO, ANDES).

[2024 – Current] **Optical Qualification of the Test Tower for the ELT's M4 Adaptive Mirror and the DP Prototype.**

Activities within the ESO E-ELT Project for the software development and calibration of the M4 Deformable Mirror (DM).

The main activities consist in:

- Software developing and debugging for the control of the Optical Test Tower designed for M4: motors for mechanical and optical alignment.
- Software developing and debugging for the DP (Demonstration Prototype) and M4 Deformable Mirrors (DMs).
- Data production and analysis for optical characterization and calibration of the M4 DM.

[Dec 2025 – Current] **Petalometer metrology bench integration**

The PetalMirror (PTL) is a 6-segments actuated mirror with 3 piezo actuators on each segment, which provide Piston, Tip and Tilt control over the segments. The PTL bench is compose of:

- The PTL;
- An Interferometer, for absolute TT measurements (and correction);

- An absolute position sensor (SPL), to measure differential piston between the actuated segments and the central, fixed, hexagonal segment, over a 6mm laser focus centered on the gap;
- An AttoCube interferometer, able to measure absolute height position of the measured area, in a 6 laser configuration, to measure all six segments.

CONFERENCES AND SEMINARS

[5 Jul 2026 – 10 Jul 2026] **SPIE Astronomical Telescopes + Instrumentation 2026** Copenhagen, Denmark

First author in the poster presentations:

- *Building a catalog of close binary stars through GAIA PSF simulations for ELT era AO systems*
- *Integration and alignment of the Petalometer testbench at INAF-Arcetri*

Co-authored talks/poster presentations:

- *Subaperture calibration of deformable mirror for ELT-era DM size*
- *Novel approach to a stitched optical surface reconstruction*
- *Optical calibration of the ELT-M4: automated testing procedure and results on the prototype*
- *Optical calibration of the ELT-M4: segment phasing procedure tested on the prototype*
- *Optical calibration of the ELT-M4: Validation of the local calibration of capacitive position sensors on the demonstration prototype*
- *SPECULA: adoption, developments, and use cases of a scalable adaptive optics simulation framework*

Link: <https://spie.org/conferences-and-exhibitions/astronomical-telescopes-and-instrumentation>

[21 Apr 2026 – 24 Apr 2026] **V ADONI Workshop** Rome, Italy

Talk about the OPTICALIB software, developed within the AO *Deformable Mirrors Group* at INAF-OAA

Link: <https://indico.ict.inaf.it/event/3479/page/1914-adoni-2026>

[27 Oct 2025 – 31 Oct 2025] **AO4ELT8** Viña del Mar, Chile

Poster presentation about the BINCAT project.

Link: <https://ao4elt8.sciencesconf.org/?lang=en>

[19 Nov 2024] **AstroBigné** INAF - Osservatorio Astrofisico di Arcetri

Short talk about my past and present research.

PRODUCTION

Publications

- N. Azzaroli, R. Briguglio, L. Oggioni, G. Pariani, C. Selmi, M. Xompero, P. Ferraiuolo, A. Riccardi; (in prep.): "*Subaperture Calibration of Deformable Mirrors for ELT-era DM size*"
- Ferraiuolo, P., Briguglio, R., Agapito, G., Véran, J.P.; (in prep.): "*Building a catalog of close binary stars through GAIA PSF simulations for ELT era AO Systems*"
- Ferraiuolo, P., et.al.; (in prep.): "*Integration and alignment of the Petalometer testbench at INAF-Arcetri*"
- Ferraiuolo, P., Briguglio, R., Xompero, M., Pariani, G., Azzaroli, N., Oggioni, L.; (in prep.): "*Optical calibration of the ELT-M4: automated testing procedure and results on the prototype*"
- Oggioni, L., Briguglio, R., Xompero, M., Pariani, G., Azzaroli, N., Ferraiuolo, P.; (in prep.): "*Optical calibration of the ELT-M4: Validation of the local calibration of capacitive position sensors on the demonstration prototype*"
- Briguglio, R., Ferraiuolo, P., Pariani, G., Xompero, M., Azzaroli, N., Oggioni, L.; (in prep.): "*Optical calibration of ELT-M4: segment phasing procedure tested on the prototype*"
- Xompero, M., Azzaroli, N., Ferraiuolo, P., Briguglio, R., Oggioni, L.; (in prep.): "*Novel approach to a stitched optical surface reconstruction*"
- Rossi, F., Agapito, G., Puglisi, T.A., et.al.; (in prep.): "*SPECULA: Adoption, Developments, and Use Cases of a Scalable Adaptive Optics Simulation Framework*"

Software - GitHub Repositories

- **OPTICALIB** python package for OPTical CALibration of deformable mirrors
- **M4** software for the control and calibration of the ESO-ELT M4DM and its calibration tower (OTT)
- **GRASP** Globular cluster Astrometry and Photometry Software
- **XuPy** Cupy interface for GPU-accelerated Masked Arrays

Link: <https://github.com/pietroferraiuolo?tab=repositories>

EDUCATION AND TRAINING

[12 Oct 2025 – 17 Oct 2025]

SCIA3

INAF - Istituto Nazionale di Astrofisica <https://andymelandri.wixsite.com/scia2025>

City: Bertinoro | **Country:** Italy | **Field(s) of study:** Scientific Communication In Astronomy 3

[21 Jul 2025 – 1 Aug 2025]

GISM3 Doctoral School

UMR Astrophysique Instrumentation Modélisation (AIM) <https://ismgalaxies2025.sciencesconf.org/>

City: Banuyls-sur Mer | **Country:** France | **Field(s) of study:** The Interstellar Medium of Nearby Galaxies

Link: <https://drive.google.com/file/d/1RR80-jmlbgTI5DI5EkzgE4iV-mH2pas/view>

[25 Feb 2025]

Course in Git, GitLab and CI/CD

CNR - Bologna & INAF <https://indico.ict.inaf.it/event/3067/>

[29 Sep 2024 – 5 Oct 2024]

AI-PHY Doctoral School

Università degli Studi di Milano - Bicocca <https://agenda.infn.it/event/40881/>

City: Monopoli (BA) | **Country:** Italy | **Field(s) of study:** Artificial Intelligence and modern PHYSics: a two-way connection | **Final grade:** Certificate of Attendance

Link: <https://drive.google.com/file/d/1nuGjUoGKD7ObFBU2cicM-DnC0NeYl-m6/view?usp=sharing>

[Jan 2021 – 23 Jan 2024]

Master's Degree in Astrophysics and Astronomy

Sapienza - University of Rome <https://www.uniroma1.it/it/pagina-strutturale/home>

Address: Piazzale Aldo Moro, 5, 00185, Roma, Italy |

Field(s) of study: Astronomy & Astrophysics ; Cosmology | **Final grade:** 110 | **Level in EQF:** EQF level 7 | **Thesis:** Dynamics of Globular Clusters: The Effective Potential through the Gaia Data Release 3

Link: <https://drive.google.com/file/d/114HtduHBJpqnzEidfCaNN9jiykxtlnmW/view?usp=sharing>

PROJECTS

[Oct 2022 – Dec 2023]

Dynamics of Globular Clusters: The Effective Potential Through the Gaia Data Release 3

Master's thesis research project, which explored the dynamics of collisional systems, in particular Globular Clusters. By analyzing GC phase-space data, their internal velocity distribution was used to extract the Effective Gravitational Potential within shells of the cluster and the cluster as a whole, providing a first observational counterpart to its theoretical derivation.

Link: https://drive.google.com/file/d/1Q-wPV1G5NUaBDUnpqOHGP_kyBJPQSKNM/view?usp=sharing

[May 2022 – Nov 2022]

Simulation of a point-like and an extended source of SZ signal and filtering optimization for MUSTANG2

Research experience provided by the master's degree, in which I had the opportunity to collaborate with Professor Elia Battistelli and his team in their project investigating the Cosmic Web through the Sunyaev-Zel'dovich effect measured in the hot gas in-between merging galaxy clusters. The project focused on simulating synthetic point-like and extended SZ sources and observing them with a CCD array similar to that of the GBT's MUSTANG-2 instrument.

Link: <https://drive.google.com/file/d/112vvZrSr0hAbSUMUNoCSjVqULpm8aPWA/view?usp=sharing>

WORK EXPERIENCE

2018-2020 Ohana Società Cooperativa Sociale Onlus | 2020-2022 Privately

City: Pomezia (RM) | Country: Italy

[Sep 2018 – Sep 2022] Learning Tutor

- Provided tutoring in scientific subjects, including mathematics and physics, to pre-teenagers and adolescents with SEN (Special Educational Needs), SLD (Specific Learning Disability), or other difficulties falling within the category of 'Learning Disorders'.
- Collaborated with a team, ensuring effective communication and management of educational support.
- Maintained regular contact with students' families, school staff, and specialized therapeutic professionals for planning and coordination.
- Contributed to team-building activities to foster a positive work environment among colleagues.

OUTREACH ACTIVITIES

[27 Sep 2024 – Current] European Bright Night for Researchers

Interacted with the general public about astrophysical themes and the what is the job of researchers .

[Apr 2024 – Current] Outreach Activities at the Observatory

Actively participating in the Observatory outreach activities, which include:

- Tour guide of the Observatory for schools and general public
- Technician and astronomer for observation nights, carried out with the MEADE telescopes.

SKILLS

Programming

Python | Git | Bash | R | C / C++ | Linux | LaTeX | Fortran90

AI Tools

Local LLMs w/ Ollama - OpenWebUI - ComfyUI | Local Agentic Workflows w/ Ollama + RooCode

Languages

English - C2 | Italian - Mother

Firenze, OAA, 13 May 2026



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